ML Flow
→ Data collection
→ Data processing
→ Apply ML model.
→ Train & Test the data.
→ Evaluate
→ Deployment

Machine learning

ML: Data + Output → p

learn to do the task bas
→ Train to do specific Task e
→ Test: how is it going to be

ML - Its a subset of AI, machines wants to learn & make decisions without explicit programming for every task.

ML is a way of teaching computers to learn from data instead of being explicitly programmed.

Eg: -

Insted of writing rule like
if age ≥18 & income >50k → approve loan.
we give the computer past loan data & it learns the patterns itself.

Eg: - Netflix recommendations
Google search

* Labeled Data - Labeled data is data that already has input + correct output.
(Data that already has the correct answer)
Input (Question)
Output (correct answer/label)

Labeled Data is used
→ Spam detection
→ Face recognition
→ Medical diagnosis
→ Fraud detection

* Unlabeled Data - Data with no correct answer
It only has input but no label

Eg:-

A bag of fruits
But no name are written
You must group them by shape or colour on our own

→ Data

Customer shopping history
Website click data
Photos with No tags

Model must
Find patterns
Create groups
Discover hidden structure

* Reinforcement learning :- (Real time learning)
→ In (RL) there is no answer key, but your RL agent still must decide how to act to perform its task.
In the absence of existing training data, the agent learns from experience.
It collects training examples (this action was good, that action was bad) through trail-and-error as it attempts its task with the goal of maximizing long-term reward.

* Supervised learning - learning without teacher.
Data is labled
Model is trained using correct answers

* How it works.
→ we give the model labeled data
→ Model learns the patterns, relationship b/w input & Output

→ We give new data to the model

→ Model predicts the answer.

* Types of Supervised learning.

→ Classification ⟶ Spam/not spam(Ham)

→ Regression ⟶ Predicting house price.

* Unsupervised learning - learning without a teacher (as in unlabeled data)
model discovers patterns on its own.

* How it works
Give the model raw data (unlabeled data)
Model looks for
Groups
Similarities
Hidden patterns
It automatically clusters the data.

Resedual or loss = the difference b/w actual value & the predicted value

* Regression - is supervised learning technique used to predict a numerical (continuous) Value.

It predicts no' not categories.

Types -

-> Simple linear Regression
predicts output using only one input feature

Formula $y = mx + b$ -> Intercept

m -> slope, x -> Input, y -> Output

Eg:- Prediction of house price using size.

-> Multiple linear Regression
predicts output using more than one input feature

Eg:- Prediction of house price using size, Bedrooms, location score.

-> Polynomial Regression - Used when data is curved not straight-line

Eg:-
Speed vs fule consumption
Growth Curves
Stock price movement.

ML-class notes.

OLS vs gradient descent- Problem eg:-

* OLS → Ordinary least Square method

Sum of squared error (SSE) = Summation $(y-y^\wedge)2$

OLS → To reduce the above value i.e the loss.

* Gradient decent- optimizer → which is used to reduce the loss

Residual sum of square

* TSS = RSS + ESS → Error sum square

Total sum square

R^2 = TSS - RSS.

How much of the model is able to identify the pattern or metric.

Adjusted R^2

R^2 will also increase by adding independent Variable → this Variable will not effect the model or the data.

OLS → is the method behind linear regression - its how the model finds the 'line of best fit' through data.

When trying to predict a target Variable 'y' using one or more features 'x'. OLS finds the coefficients (Slope & Intercept) that make this equation fit best

Best fit → meaning the sum of squared residuals.

Residual → the gap b/w what the model predicted & the actual Value.

Gradient descent → is how most ML models actually find their best parameters when there's no clean formula for them directly.

standing on a hill hieigh = error (how wrong your model is). You want to walk downhill to the lowest point -- that's the minimum error.

TSS → measures how much your target Variable Varies on its own before any model gets involved its the baseline your model has to beat.

$TSS = ESS + RSS$

RSS → Residual Sum of Square → the error your model couldn't explain

ESS → Explained Sum of square → The error your model successfully accounted for relative to guessing the mean.

R^2 → How much better is my model than just guessing the average.

Eg:- Score from 0-1 → that tells you how well your model explains the ups & down's in your data.

$R^2 = 0$ → Your model is no better than Just predicting the average

$R^2 = 1$ → Your model predicts Perfectly, no errors at all

$R^2 = 0.75$ → Your model explains 75% of the Variation in the data, the remaining 25% is unexplained noise/error

Adjusted

→ ad
into the
improv
from

Add

Add

Adjusted R^2 → but does it actually deserve credit for those extra features?

→ adds a penalty for every feature you throw into the model. It only goes up if the new feature improves the fit by more that what you'd expect from random chance.

Add a genuinely useful feature → Adjusted R^2 goes up

Add a useless/Random feature → Adjusted R^2 goes down.